

# Depositor Discipline in Community Banking

Evidence from the CECL Information Shock

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Dimuthu Ratnadiwakara

Federal Reserve Bank of Richmond

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# Do uninsured depositors actually discipline banks?

Theory says they should; causal evidence has been elusive

- ▶ Monitoring by uninsured claimants is the core private check on bank risk-taking  
Calomiris–Kahn '91; Diamond–Rajan '01
- ▶ In the data, riskier banks pay higher rates and hold fewer uninsured deposits: depositors decline to roll over maturing claims Park–Peristiani '98; Martinez Peria–Schmukler '01; Goldberg–Hudgins '02
- ▶ **Policy stakes are current:** Main Street Depositor Protection Act would insure \$10M per *business* account; reciprocal networks, sweeps, and brokers already convert uninsured money into insured money Kim–Kundu–Purnanandam '24; Huberdeau–Reid–Pennacchi '25

But is that correlation **discipline** (depositors reacting to news about risk) or **equilibrium sorting** (risky banks and yield-chasing depositors matching)?

⇒ **Needed: a shock to information about bank risk that is unrelated to concurrent bank fundamentals**

# Why the causal evidence has been elusive: three problems

- ▶ **Simultaneity.** A bank's risk profile, deposit mix, and depositor behavior co-evolve. Regressions of flows on risk measures capture sorting, not a disciplinary response
- ▶ **Accounting opacity.** Under the incurred-loss standard banks could delay recognition and understate risk in disclosures, so measured risk was partly a reporting choice Beatty–Liao '11; Bushman–Williams '15
- ▶ **Aggregate stress.** Episodes with the most variation in bank distress are also the episodes with expanded guarantees, system-wide shocks, and the highest liquidity needs of depositors themselves. All three mute or contaminate the depositor response Berger–Turk-Ariss '15; Bennett–Hwa–Kwast '15

⇒ **The mandatory CECL adoption of 2023 helps me address these endogeneity concerns**

# What is CECL? Lifetime expected losses replace incurred losses

ASU 2016-13 (FASB 2016), codified in ASC 326. The largest change to bank loan accounting in decades

## Before: incurred loss (ILM)

recognize a loss only when *probable and estimable*  
⇒ provisions lag defaults, disclosures understate credit risk

## After: expected loss (CECL)

reserve for *lifetime* expected credit losses at origination and every reporting date ⇒ credit risk forced into a reported number

At transition, the entire existing loan book is re-reserved on a single day:

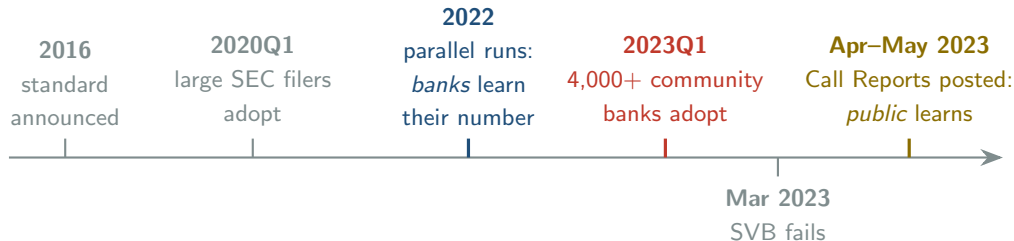
$$\Delta_i = ALLL_{i,t^*}^{CECL} - ALLL_{i,t^*-1}^{ILM} \quad (\text{one-time charge to retained earnings, no cash flow})$$

$$\text{Treatment:} \quad CECL \text{ Adj}_i = \frac{\Delta_i}{Equity_{i,t^*-1}} \times 100$$

⇒  $\Delta_i$  reflects the composition and seasoning of loans made years earlier, not deposit-market conditions in 2023

# The experiment: a mandatory disclosure with a predetermined size

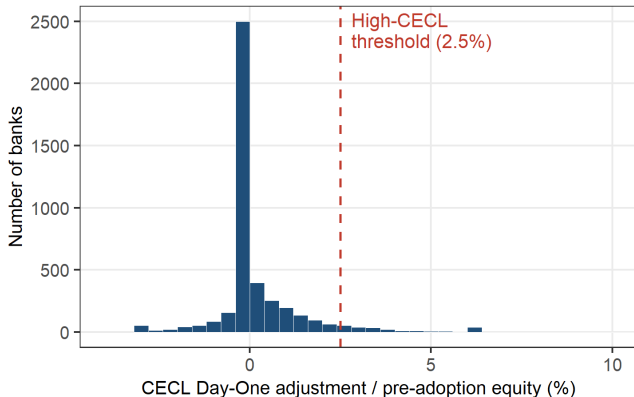
CECL Day-One adjustment = one-time charge to retained earnings, fixed by past underwriting



- ▶ Pure accounting event: no cash flow, no change in loan risk. Only **news** about lifetime expected losses already embedded in the book
- ▶ The adoption date was public years in advance. The size of each bank's charge was not observable outside the bank until the Call Report was posted
- ▶ Timing wedge: banks knew their number in 2022, the public only in April–May 2023

# Treatment: the disclosed charge, scaled by equity

4,320 community banks (<\$10B); charge disclosed on Schedule RI-A of the adoption-quarter Call Report



**92%** adopt in 2023Q1

~**half** disclose a charge of about zero

**217** High-CECL banks: charge  $\geq 2.5\%$  of equity ( $\approx 95$ th pctile), mean 4% of equity

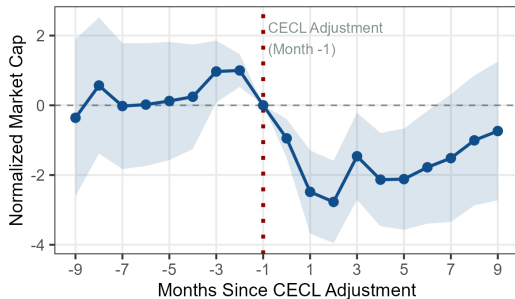
**Charge is hard to predict:** 2022Q4 observables explain  $\sim 1\%$  of its variation

[▶ detail](#)

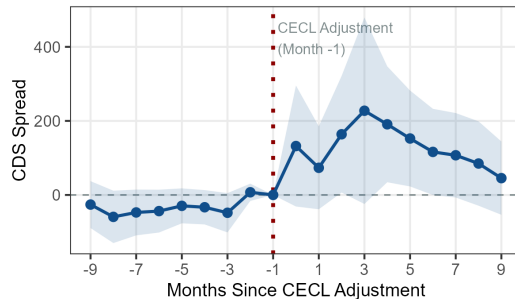
# The Day-One number is news: markets repriced it in 2020

Large SEC filers adopted 2020Q1, three years before community banks; monthly event studies, coefficient on charge/equity

Market capitalization



5-year CDS spread



⇒ Equity falls ~2.8% per pp of charge at adoption; CDS spreads widen with a lag. Sophisticated investors treat the number as material

# Data: quarterly Call Reports for 4,320 community banks, 2016–2025

FFIEC 041/051, 2016Q1–2025Q4 (~97,000 bank-quarters); RateWatch posted rates; CRSP + Bloomberg for the 2020 cohort

- ▶ Federally insured banks < \$10B in assets at 2022Q4; median bank \$0.3B
- ▶ Deposit breakdowns by insurance status from Schedule RC-E; outcomes scaled by assets
- ▶ Time deposits fund **12.7%** of assets (5.3 uninsured + 7.4 insured)

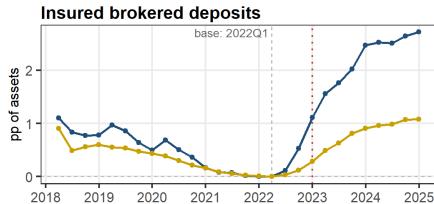
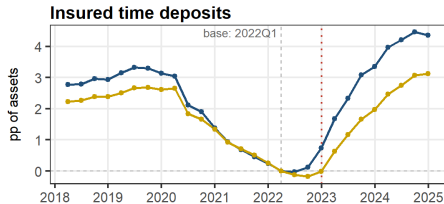
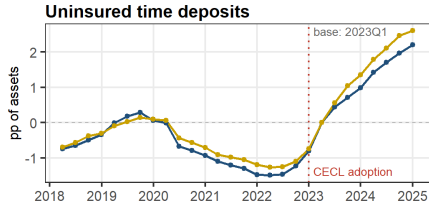
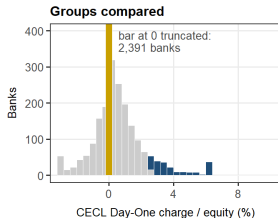
| Pre-adoption means            | High CECL | Low CECL | Hi–Lo    |
|-------------------------------|-----------|----------|----------|
| CECL Adj / Equity (%)         | 3.99      | 0.08     | 3.91***  |
| NPL / Loans (%)               | 0.60      | 0.45     | 0.15***  |
| Equity / Assets (%)           | 8.88      | 10.34    | –1.46*** |
| Ins. time dep. / Assets (%)   | 9.13      | 7.27     | 1.86***  |
| Unins. time dep. / Assets (%) | 5.62      | 5.28     | 0.33     |

High-CECL banks: weaker loan quality, thinner capital. But **balanced on the primary outcome** (bank FE absorb the levels)



# Univariate evidence: uninsured money leaves, insured money arrives

Group means, pp of assets; bases 2023Q1 (top) and 2022Q1 (bottom); 206 High-CECL vs 2,265 near-zero banks; no controls or FE



— High CECL — Near-zero CECL

# Empirical strategy: two specifications

Bank + calendar-quarter FE; SEs clustered by bank; controls: lagged log assets, equity/assets, int. exp./assets

## Specification 1: level DiD across banks (expected sign $\beta < 0$ )

$$y_{it} = \alpha_i + \gamma_t + \beta \text{Post}_{it} \times \text{CECL}_i + \mathbf{X}'_{it-1} \delta + \varepsilon_{it}$$

$\text{CECL}_i$ : continuous CECL Adj or High-CECL indicator; event studies replace Post with event-time dummies,  $k = -1$  omitted. Quarter FE absorb the tightening cycle and any aggregate 2023-stress response.

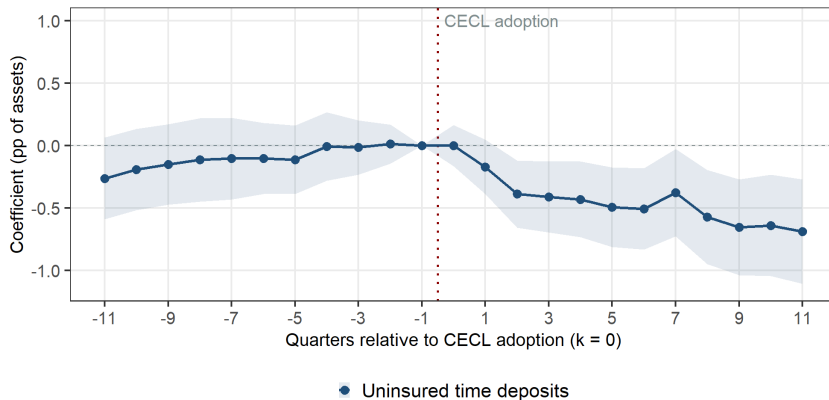
## Specification 2: triple difference within bank $\times$ quarter (series stacked)

$$y_{idt} = \alpha_{it} + \beta_2 \text{Post}_t \times \text{Unins}_d + \beta_3 \text{CECL}_i \times \text{Unins}_d \\ + \beta_4 \text{CECL}_i \times \text{Post}_t \times \text{Unins}_d + \varepsilon_{idt}$$

$\alpha_{it}$  absorbs every bank-level shock in every quarter: funding demand, stress exposure, liability strategy.  $\beta_4$  is identified only from the within-bank uninsured-vs-insured differential.

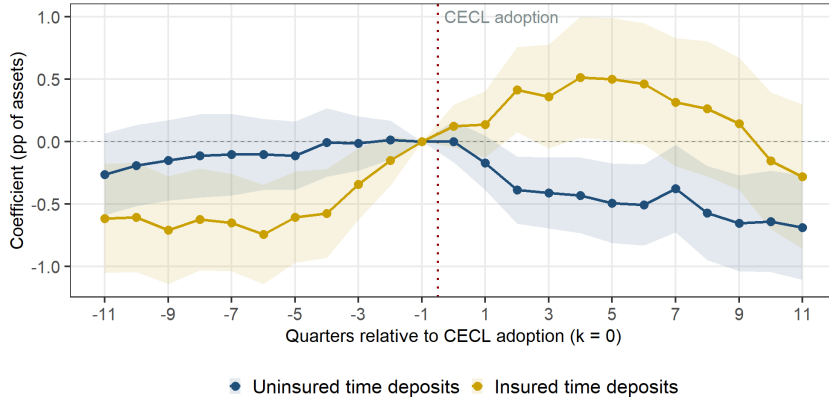
# Uninsured time deposits are flat for ten quarters... then break at disclosure

Event study, High-CECL  $\times$  event time; bank + quarter FE; 90% CI



## ...and insured time deposits rise to meet them

Same specification, insured series overlaid; identical axes



# Magnitudes: a composition shift, not a funding collapse

Coefficients in pp of assets; bank-clustered inference; \*/\*\*/\*\* = 10/5/1%

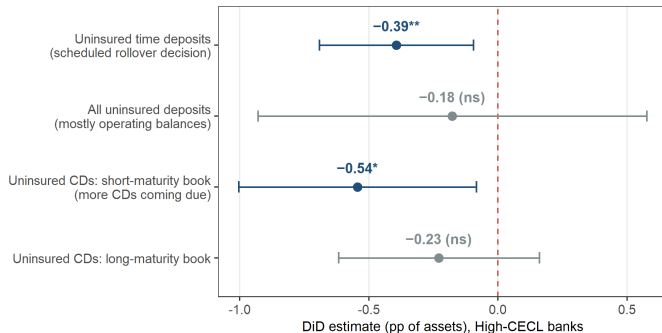
► specification

|                              | Unins. time dep.<br>/ assets | Ins. time dep.<br>/ assets | Unins. – ins. gap<br>(composition) |
|------------------------------|------------------------------|----------------------------|------------------------------------|
| Post × High CECL             | −0.39**                      | +0.71**                    | −1.25***                           |
| Post × CECL Adj (continuous) | −0.072**                     | +0.070                     | −0.177***                          |
| Bank + quarter FE            | ✓                            | ✓                          | —                                  |
| Bank × quarter FE            | —                            | —                          | ✓                                  |
| Observations                 | 97,064                       | 97,064                     | 194,128                            |

**The boxed cell is the headline estimate:** Post × High CECL on uninsured time deposits/assets, bank + quarter FE. The next slides take this one specification apart: by deposit type, by capital, by maturity, and by where the money went.

# Discipline lives exactly where the depositor holds a decision

Same specification as the table. Interaction coefficients shown graphically with 90% CIs from here on



## Why no response in total uninsured deposits?

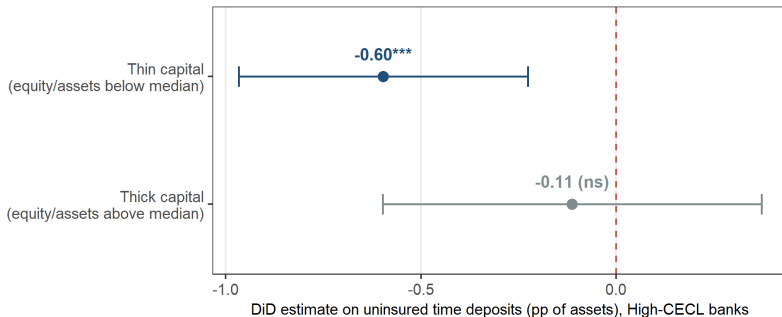
At the median bank, balances over \$250k are 34% of assets. Of these, 23% are municipal deposits, collateralized by pledged securities, and 26% are noninterest-bearing operating accounts, held for liquidity services rather than yield d'Avernas et al. '23.

This money has no scheduled decision point and no reason to respond to a credit disclosure.

⇒ **No response in operating balances; twice the response where more CDs came due. A quantity response at the rollover margin**

# The response is concentrated at thinly capitalized banks

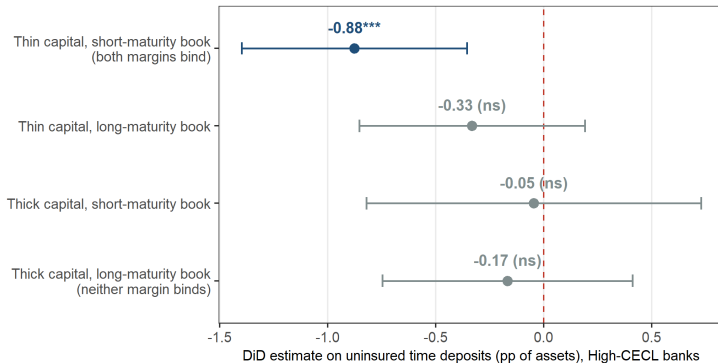
Post  $\times$  High CECL on uninsured time deposits/assets, split at median 2022Q4 equity/assets (9.8%); bank + quarter FE; 90% CI [▶ continuous version](#)



⇒ Same disclosed charge, no response in the better-capitalized half. A given charge is worse news when the buffer absorbing it is thin

## Both margins together: thin capital and a scheduled exit

Post  $\times$  High CECL on uninsured time deposits/assets, in the  $2 \times 2$  of 2022Q4 capital (median 9.8%) and 2022Q4 share of uninsured CDs maturing within one year (median 76%); bank + quarter FE; 90% CI

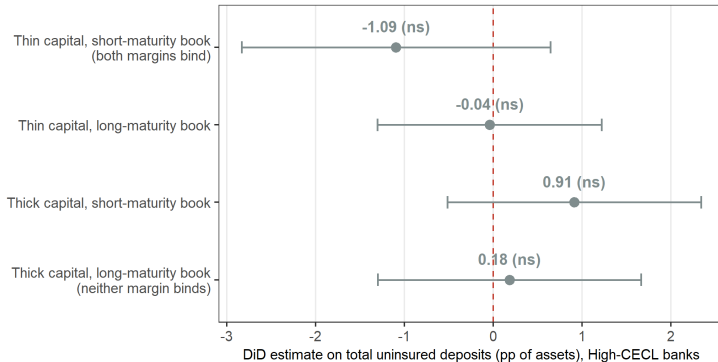


⇒ The response sits in one cell: banks where the charge was bad news and depositors had a near-term decision to make



# The same 2×2 for total uninsured deposits

Post × High CECL on total uninsured deposits/assets (mean 35% of assets); same cells, controls, and fixed effects as the previous slide; 90% CI

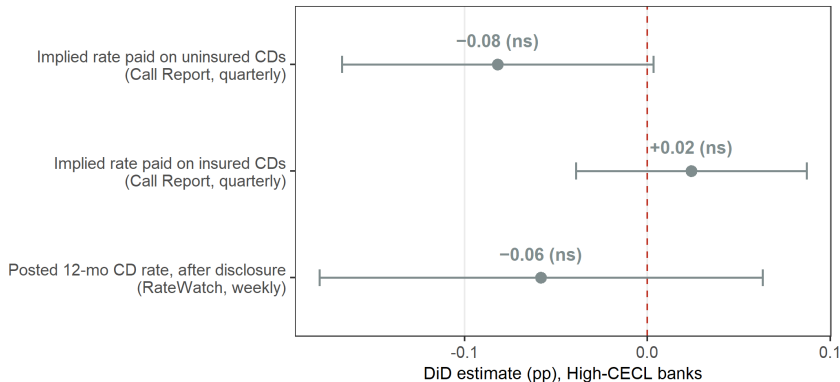


⇒ Same sign in the same cell, but the aggregate is seven times larger and mostly operating money: the CD signal is lost in the noise

# High-CECL banks never defended their funding on price

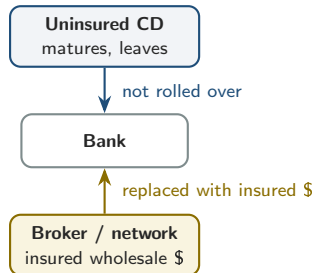
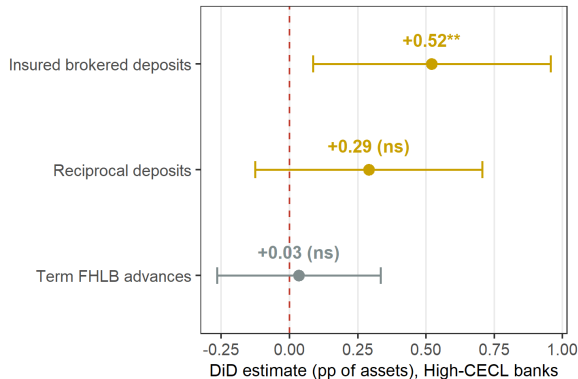
Implied rates from Call Report interest expense; posted APYs from RateWatch, bank-week panel

[▶ detail](#)



# Where the money went: banks repurchased the lost insurance

DiD estimates for the three replacement margins, High-CECL banks, pp of assets

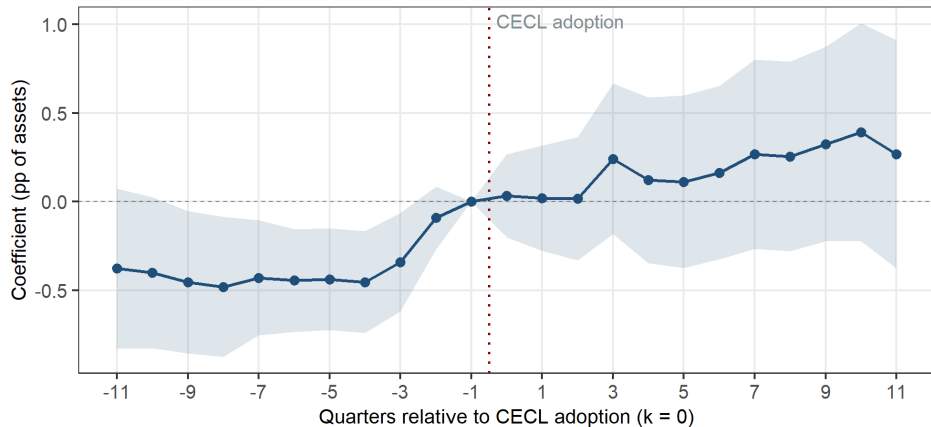


Total funding is unchanged. What changes is the *insurance status* of the funding. No move in uninsured brokered money or FHLB borrowing.

# Banks moved first, on private information

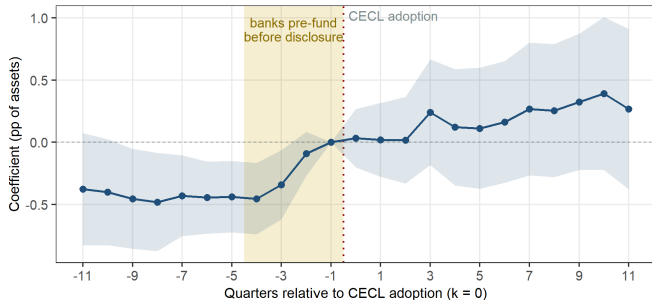
Insured brokered deposits, event study; High-CECL  $\times$  event time; 90% CI

► phase-in mechanics



# Banks moved first, on private information

Banks ran CECL in parallel through 2022. Management knew the number before the public did



**The ramp scales with the still-undisclosed charge.**

$\Delta$  insured brokered/assets,  $k = -4$  to  $-1$ , on the charge disclosed at  $k = 0$ :

+0.13\*\*\* pp per pp of charge

+0.53\*\*\* pp for High-CECL banks

**And does not vary by capital or maturity.**

Interactions with 2022Q4 capital and CD maturity all insignificant. The depositor response sits in one cell; the bank response does not.

Phase-in electors pre-funded 0.41\*\* pp

more.

[▶ detail](#)

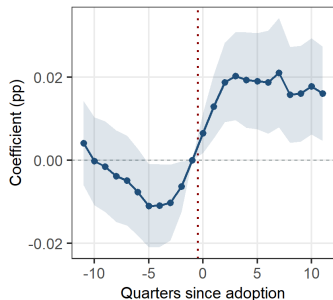
[▶ who kept it](#)

⇒ **Bank margins move on private information, before adoption; depositor margins move only when the number becomes public**

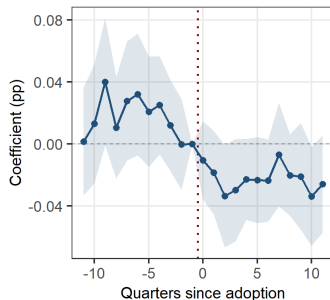
# The substitution is costly: funding costs up, profitability down

Event studies for interest expense, ROA, NIM; High-CECL  $\times$  event time; 90% CI

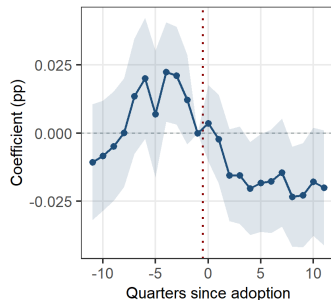
Interest expense  $\uparrow$  8–9bp/yr



ROA  $\downarrow$  15bp/yr



NIM  $\downarrow$  9bp/yr



$\Rightarrow$  No premium to stayers, no higher posted rates. The cost comes from replacing maturing uninsured CDs with insured wholesale money raised at wholesale rates

# Is this the SVB stress in disguise?

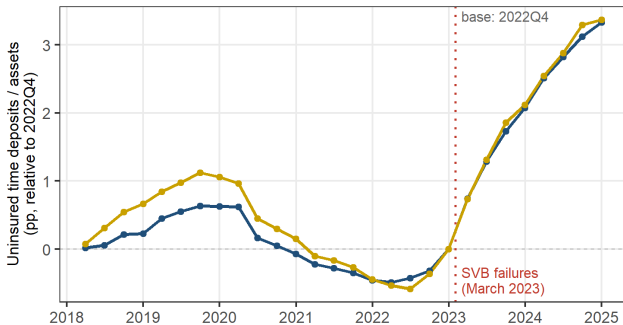
CECL disclosures reached the public in April–May 2023, weeks after the March 2023 failures

- ▶ The concern: uninsured depositors were already running from vulnerable banks. The CECL charge may simply proxy for that vulnerability
- ▶ Three answers, in turn:
  1. **Placebo.** Banks with no CECL news, sorted on run exposure: do their uninsured CDs separate after March 2023?
  2. **Orthogonality.** Is the charge correlated with unrealized losses, asset maturity, or uninsured funding?
  3. **Horse race.** Do the CECL estimates survive controlling for stress exposure directly?

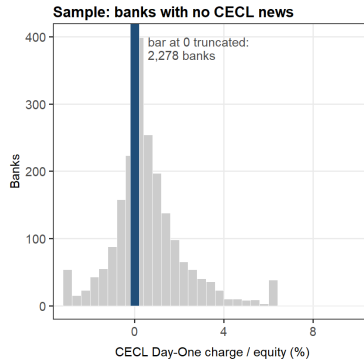
# Placebo: the same picture, drawn on stress exposure, is flat

Zero-charge banks split at the median 2022Q4 unrealized loss; group means, no controls or FE

▶ placebo detail



● High unrealized loss ● Low unrealized loss



■ Zero charge (placebo sample) ■ Other

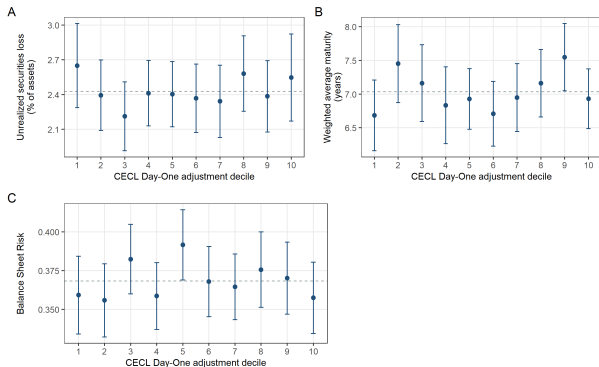
⇒ Banks with no CECL news show no separation on run exposure



# Treatment is orthogonal to run exposure

2022Q4 stress-vulnerability characteristics across deciles of the disclosed charge (positive-adjustment banks)

► placebo detail



- Treatment  $\perp$  run exposure: unrealized losses, asset maturity, Balance Sheet Risk Huberdeau-Reid-Pennacchi '25 all flat; <1% of variation explained
- **Placebo:** at zero-charge banks, uninsured CDs did *not* fall with unrealized losses after March 2023 (wrong sign: +0.07)
- Horse race: CECL estimates unchanged
- 2023 runs were wholesale, at public banks; this sample is private and retail-funded Cipriani et al. '24

# What I cannot rule out: attention

The design identifies the response to the disclosure, conditional on the level of depositor attention in 2023

- ▶ Treatment is not stress exposure. But 2023 was a year of unusual attention to bank fragility
- ▶ Depositors who had just watched two banks fail may have read the disclosure more carefully
- ▶ If so, the estimates are an upper bound on what the same number would do in a quiet year
- ▶ Banks pre-funded in 2022, before SVB, in proportion to their charge. They expected depositors to react before there was any run to watch

# Takeaways

A mandatory credit-risk disclosure moved uninsured funding at solvent community banks

1. **Depositors read the number and acted.** Uninsured CDs fell and insured CDs rose in proportion to the disclosed charge. Flat for five years before, breaking only at public disclosure
2. **Discipline = quantity exit at the rollover margin.** No repricing, no move in operating balances, stronger where more CDs came due and where capital buffers were thin
3. **Banks repurchased the insurance.** Insured brokered money replaced monitored money; interest expense rose and profitability fell. The bite of discipline is a price, paid by the bank
4. **Private coverage may already be dampening discipline.** Reciprocal networks, sweeps and insured brokered CDs convert uninsured balances into insured ones with no change in the balance sheet. The monitored share of funding shrinks before any statutory change
5. **Policy:** the discipline documented here lives in time deposits, not in operating accounts. Proposals that insure \$10M business accounts target the balances that do not monitor; the relevant margin is whether coverage reaches the CD book

## Backup: the charge is hard to predict

Cross-section of the Day-One charge on 2022Q4 observables

- ▶ Loads modestly on NPL ratio, size, capital, deteriorating ROA trend in directions consistent with credit-sensitive books needing larger lifetime reserves
- ▶ Adjusted  $R^2 \approx 1.4\%$  in the most saturated specification (fundamentals + deposit mix + performance trends + local market HHI)
- ▶ High-CECL banks are *not* observably weak banks relabeled: almost all variation is orthogonal to anything a depositor could see before disclosure

## Backup: rate tests in detail

Implied rates: Call Report interest expense on time deposits above/below \$250k (2017–)

- ▶ Implied rate on uninsured CDs, Post  $\times$  High-CECL:  $-0.082$  (SE 0.052). No premium extracted by depositors who stayed
- ▶ Within-bank large–small rate spread:  $-0.110^*$ ; if anything narrows
- ▶ RateWatch posted APYs (weekly,  $\sim 3,300$  banks): 12-mo CD \$10k/\$100k tiers and savings, no differential move in stress window ( $-0.026$ , SE 0.048) or post-publication ( $-0.058$ , SE 0.074)
- ▶  $>90\%$  of branches follow centrally set rates Begenau–Stafford '23; healthy banks can bid for uninsured money, weak banks cannot Maechler–McDill '06

## Backup: the capital phase-in election

Call Report Schedule RC-R Part I item 2.a; elected on the first CECL filing, irrevocable

- ▶ Election spreads the Day-One hit to *regulatory* capital over three years: 75% added back in year one, 50%, 25%, full impact from year four
- ▶ Economically sensible only if the bank expects a material hit; election probability rises 0.072\*\*\* per pp of charge
- ▶ 34% of High-CECL banks elect vs. 6% of the rest; electors' pre-adoption insured-brokered ramp is 0.41\*\* pp larger
- ▶ Election is measured *at* the first CECL filing; only the brokered ramp is genuinely quarters early

## Backup: who kept the brokered money?

High-CECL banks that pre-funded (insured brokered ramp  $> 0$ ,  $k = -4$  to  $-1$ ), split at the median post-adoption change in uninsured CDs ( $k = -1$  to  $+8$ ); pp of assets, winsorized 1/99

|                                 | N  | Pre-adoption ramp | Post $\Delta$ brokered | $\Delta$ unins. CDs |
|---------------------------------|----|-------------------|------------------------|---------------------|
| Uninsured CDs grew least        | 44 | +4.16             | +2.46                  | +0.53               |
| Uninsured CDs grew normally     | 16 | +2.95             | -1.22                  | +5.05               |
| Difference in $\Delta$ brokered |    |                   | +4.33*** (1.34)        |                     |

- ▶ Both groups bought 3–4 pp of insured brokered money before the number was public
- ▶ Where uninsured CDs then failed to grow, the brokered money stayed and grew; where they grew normally, it ran off
- ▶ Difference controls for ramp size, log assets, equity/assets (robust SE)

Descriptive: 16 banks in the second row; split is on a post-adoption outcome. Uninsured CDs rose at almost every bank in 2023–25, so “grew least” is relative, not a net outflow.

## Backup: capital-buffer interaction, full specification

Moderator: equity/assets frozen at 2022Q4, winsorized 1/99; gap outcome = uninsured – insured time deposits/assets

- ▶  $\text{Post} \times \text{CECL Adj} \times \text{Equity}_{22Q4}$ :  $+0.0665^{***}$  (0.019); binary analog  $+0.31^{***}$
- ▶ Main effect at zero equity:  $-0.80^{***}$  (continuous),  $-4.12^{***}$  (binary)
- ▶ Level-outcome analog:  $+0.0275^{**}$ ; robust to dropping the time-varying equity control
- ▶ Implied effect crosses zero at equity  $\approx 12\text{--}13\%$ , just above the 75th percentile; best-capitalized quartile shows essentially no response



## Backup: SVB placebo and horse race

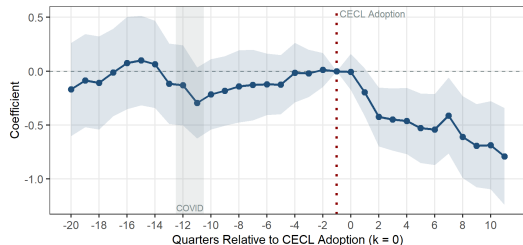
DiD around March 2023 with 2022Q4 unrealized securities loss as intensity

- ▶ Zero-charge banks ( $\sim 2,278$ , no CECL news):  $\text{Post-SVB} \times \text{loss} = +0.073^{**}$  for uninsured CDs, the *wrong sign* for a run; insured CDs move the same way ( $+0.062^*$ )  $\Rightarrow$  rate-cycle funding mix, not flight
- ▶ Full-sample horse race: CECL coefficients essentially unchanged ( $-0.078^{**}$  continuous,  $-0.406^{**}$  High-CECL); loss interaction small and insignificant
- ▶ Uninsured deposit share enters the charge cross-section *negatively* ( $-0.003^{**}$ ): high-CECL banks held slightly fewer of the deposits that ran in 2023

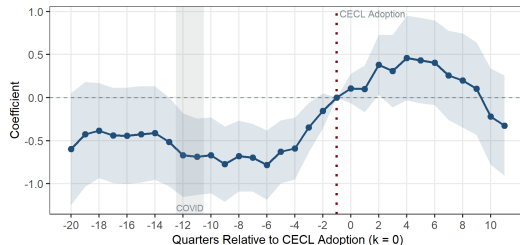
# Backup: five-year pre-period, balanced endpoints

Extended event studies from 2018Q1; edge bins pool  $k \leq -20$  and  $k \geq +11$ ; joint pre-trend tests

Uninsured time deposits ( $p = 0.18$ )



Insured time deposits ( $p = 0.37$ )



Late upward drift in the insured series is the banks' own pre-funding footprint (insured brokered CDs are booked as time deposits); absorbed into the  $k = -1$  baseline, so post coefficients *understate* the total movement.